Finding 1..n numbers sum and avg [mean].

1 to 5 number sum = 15

15/5=3 is avg

Find 1..n even, odd numbers sum

5 < 1+3+5 = 9 ✓

2+4 = 6 ✓

TC - exit

File Edit Run Compile Project Options Debug Break/watch

Line 13 Col 44 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n, s=0;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)
{
s+=i;
i++;
}
printf("Sum=%d, Avg=%.2f", s, (float)s/n); /* Type casting */
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

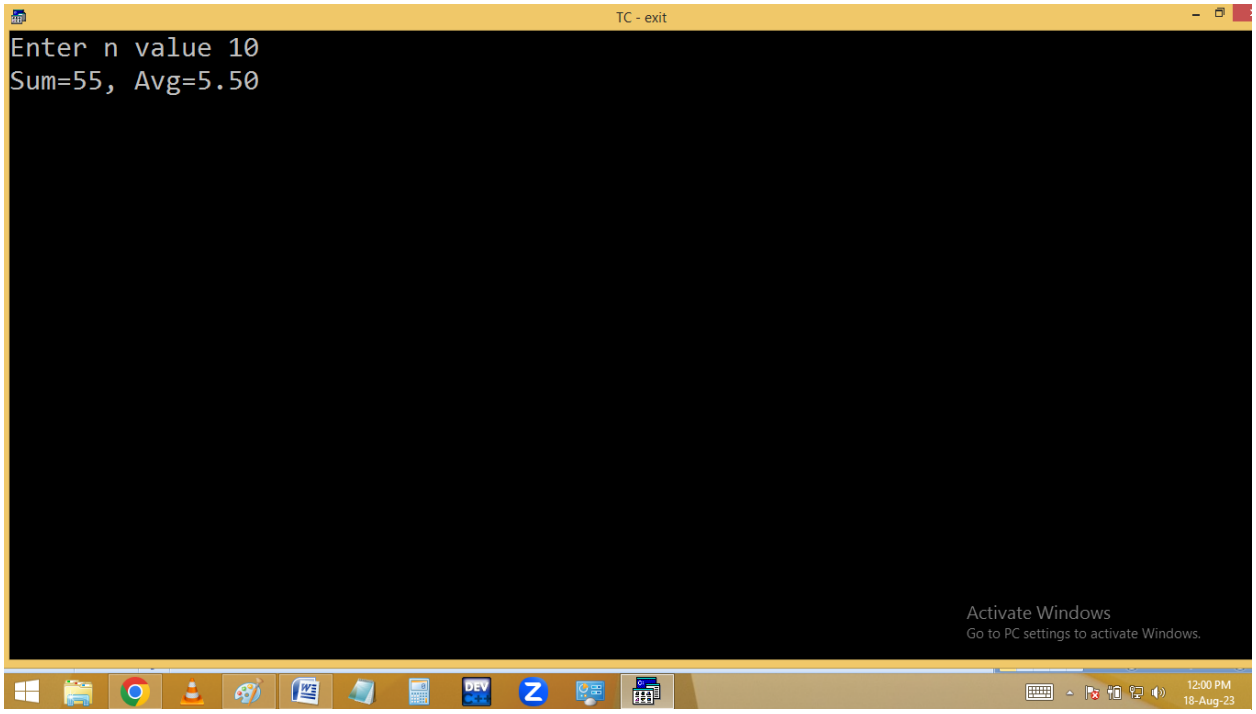
11:22 AM
18-Aug-23

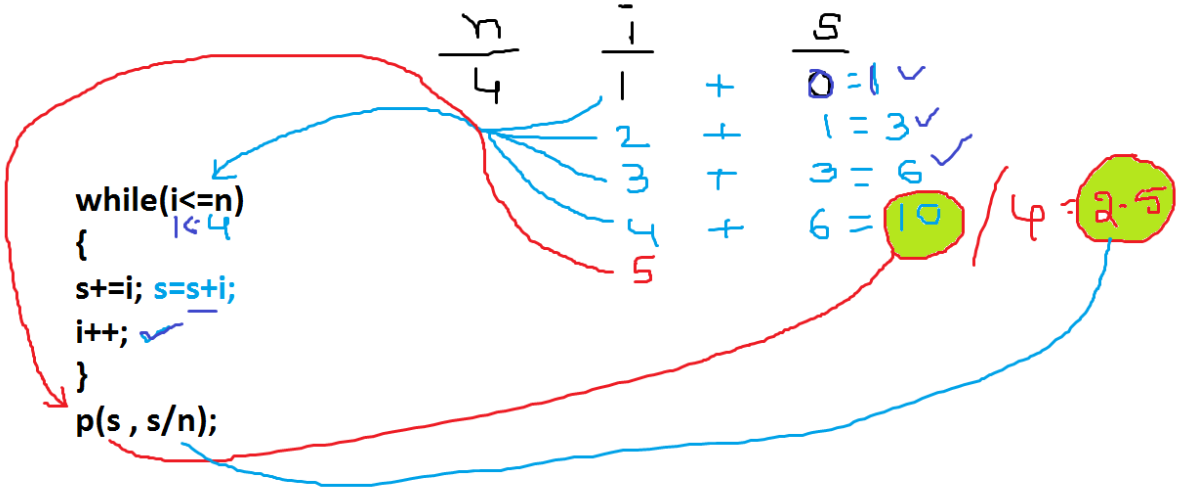
TC - exit

```
Enter n value 4
Sum=10, Avg=2.50
```

Activate Windows
Go to PC settings to activate Windows.

11:59 AM
18-Aug-23





Without using a loop:

TC - exit

File Edit Run Compile Project Options Debug Break/watch

Line 5 Col 10 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  n, s;
clrscr();
printf("Enter n value "); scanf("%d",&n);
s=n*(n+1)/2;
printf("Sum=%d, Avg=%.2f", s, (float)s/n); /* Type casting */
getch();
}
```

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TC - exit

```
Enter n value 4
Sum=10, Avg=2.50_
```

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$$\frac{n}{4}$$

s = n*(n+1)/2;

s = 4*5/2=10

Eg. Finding factorial value.

eg: 4 factorial is 4*3*2*1=24

TC - exit

File Edit Run Compile Project Options Debug Break/watch

Line 13 Col 20 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  n, f=1;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>=1)
{
f=f*n;
n--;
}
printf("Factorial=%d",f);
getch();
}
```

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TC - exit

```
Enter n value 4
Factorial=24_
```

Activate Windows
Go to PC settings to activate Windows.

TC - exit

```
Enter n value 7
Factorial=5040_
```

Activate Windows
Go to PC settings to activate Windows.

12:07 PM
18-Aug-23

TC - exit

```
Enter n value 8
Factorial=-25216
```

Activate Windows
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12:08 PM
18-Aug-23

TC - exit

File Edit Run Compile Project Options Debug Break/watch

Line 14 Col 21 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  n;
long int f=1;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>=1)
{
f=f*n;
n--;
}
printf("Factorial=%ld",f);
getch();
}
```

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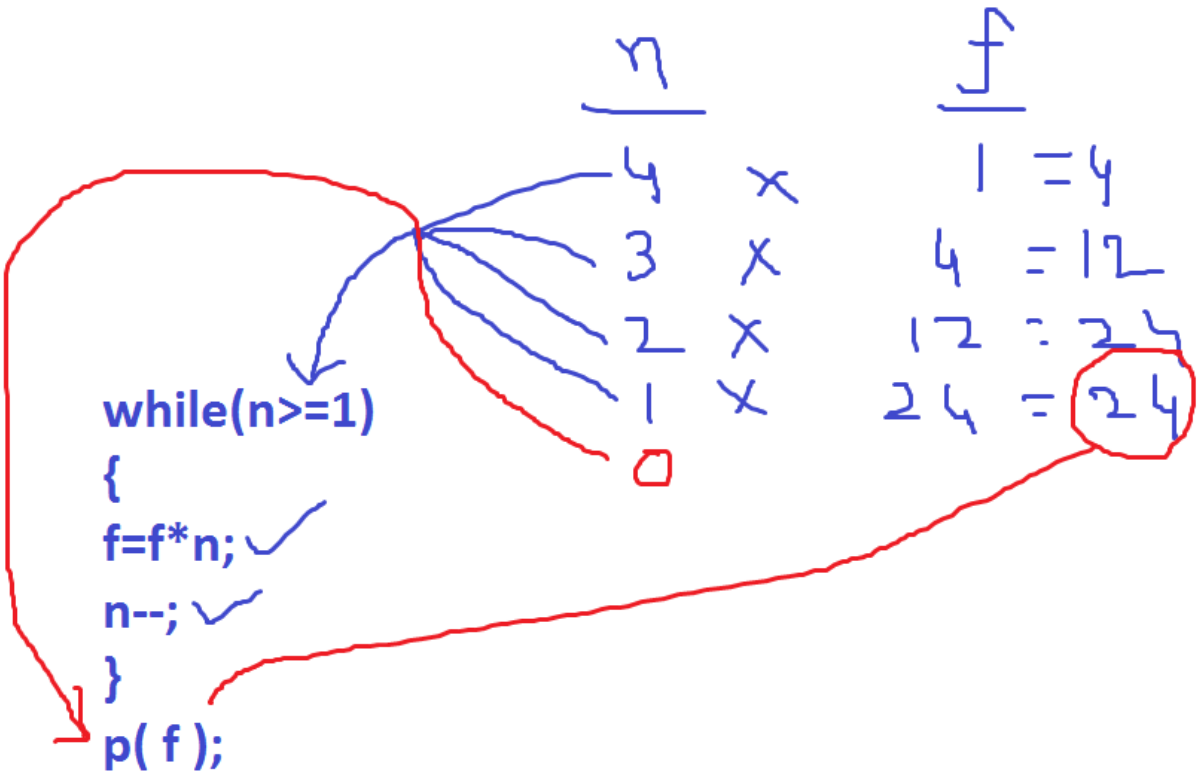
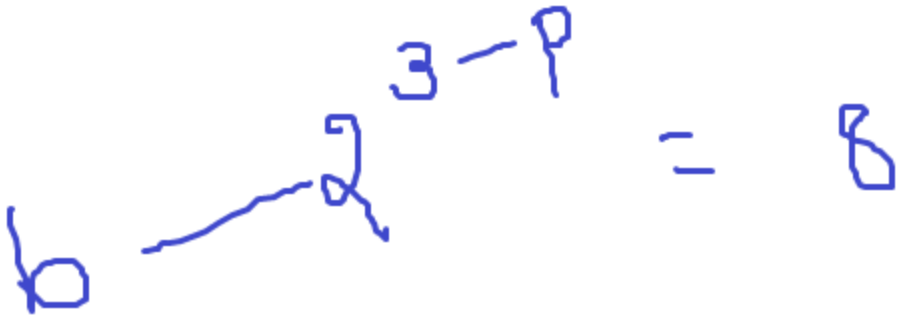
TC - exit

```
Enter n value 8
Factorial=40320_
```

Activate Windows
Go to PC settings to activate Windows.



Finding power using a user defined program:



2 *2 * 2 = 8

TC - exit

File Edit Run Compile Project Options Debug Break/watch

Line 14 Col 23 Insert Indent Tab Fill Unindent * E:11AM.C

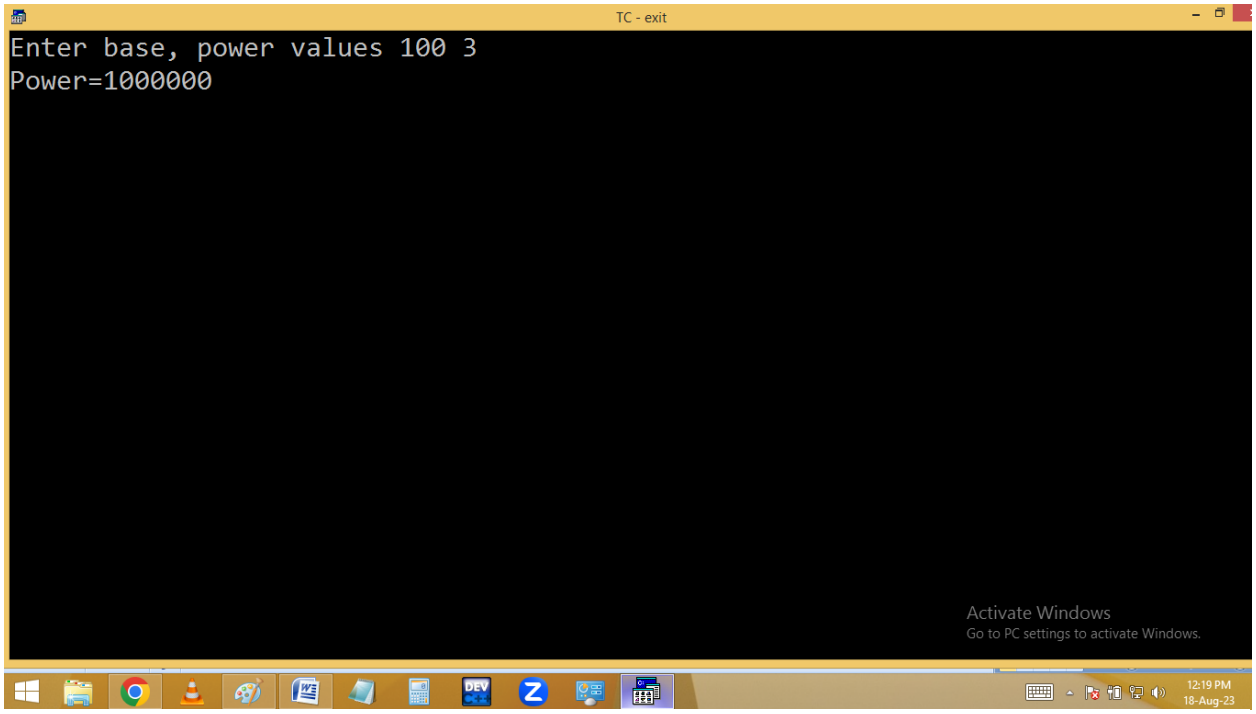
```
#include<stdio.h>
#include<conio.h>
void main()
{
int  b,p;
long pwr=1;
clrscr();
printf("Enter base, power values "); scanf("%d %d",&b, &p);
while(p>=1)
{
pwr=pwr*b;
p--;
}
printf("Power=%ld",pwr);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC - exit

```
Enter base, power values 2 3
Power=8_
```

Activate Windows
Go to PC settings to activate Windows.




```
while(p>=1)
{
pwr = b * pwr;
p--;
}
p(pwr);
```

p	b		pwr
3	2	x	1=2
2	2	x	2=4
1	2	x	4=8
0			

3 = 1

2 x 2 x 2 = 8

TC - exit

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Line 15 Col 67 Insert Indent Tab Fill Unindent * E:11AM.C

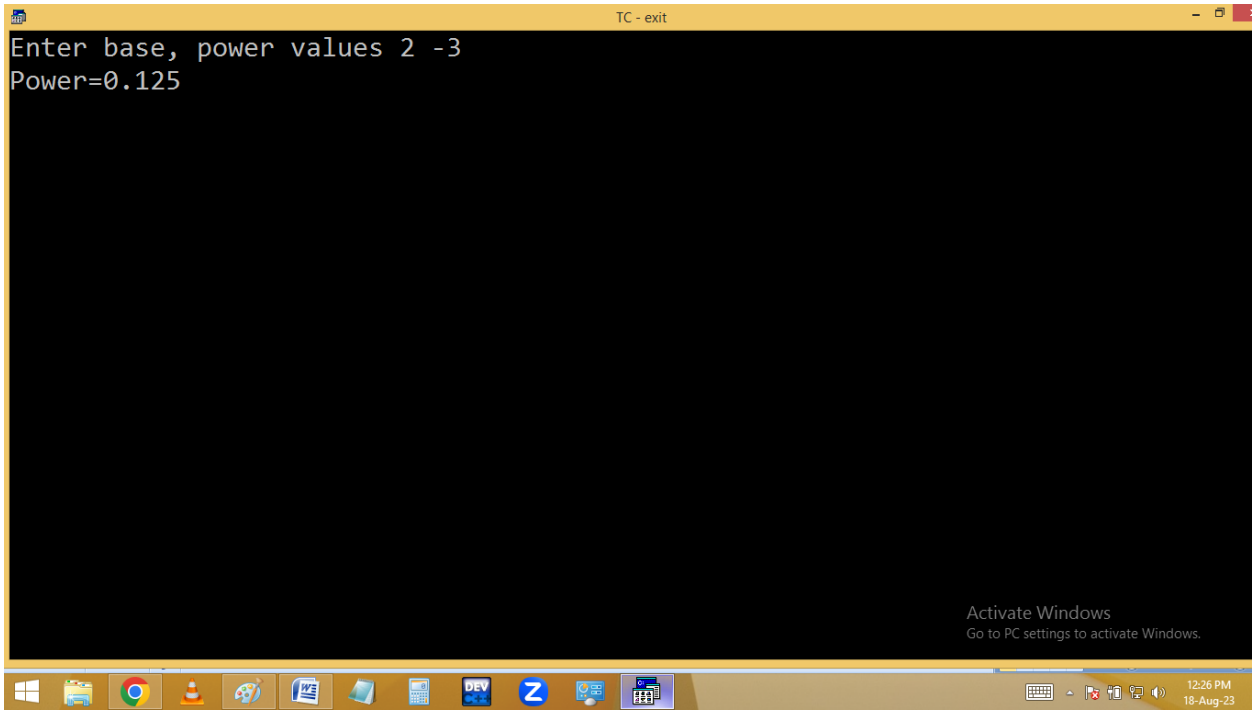
```
#include<stdio.h>
#include<conio.h>
void main()
{
int  b,p,t;
long pwr=1;
clrscr();
printf("Enter base, power values "); scanf("%d %d",&b, &p); t=p;
if(p<0)p=-p;
while(p>=1)
{
pwr=pwr*b;
p--;
}
if(t>=0)printf("Power=%ld",pwr);else printf("Power=%.3f",1.0/pwr);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC - exit

```
Enter base, power values 2 3
Power=8_
```

Activate Windows
Go to PC settings to activate Windows.



Finding powers sum.

$2^3 = 8$

$2^3 = 2^1 + 2^2 + 2^3 = 2 + 4 + 8 = 14$

TC - exit

File Edit Run Compile Project Options Debug Break/watch

Line 14 Col 29 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  b,p;
long pwr=1,s=0;
clrscr();
printf("Enter base, power values "); scanf("%d %d",&b, &p);
while(p>=1)
{
pwr=pwr*b;s+=pwr;
p--;
}
printf("Powers sum=%ld", s);_
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC - exit

```
Enter base, power values 2 3
Powers sum=14_
```

Activate Windows
Go to PC settings to activate Windows.

Using pow():

TC - exit

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Line 15 Col 20 Insert Indent Tab Fill Unindent * E:11AM.C

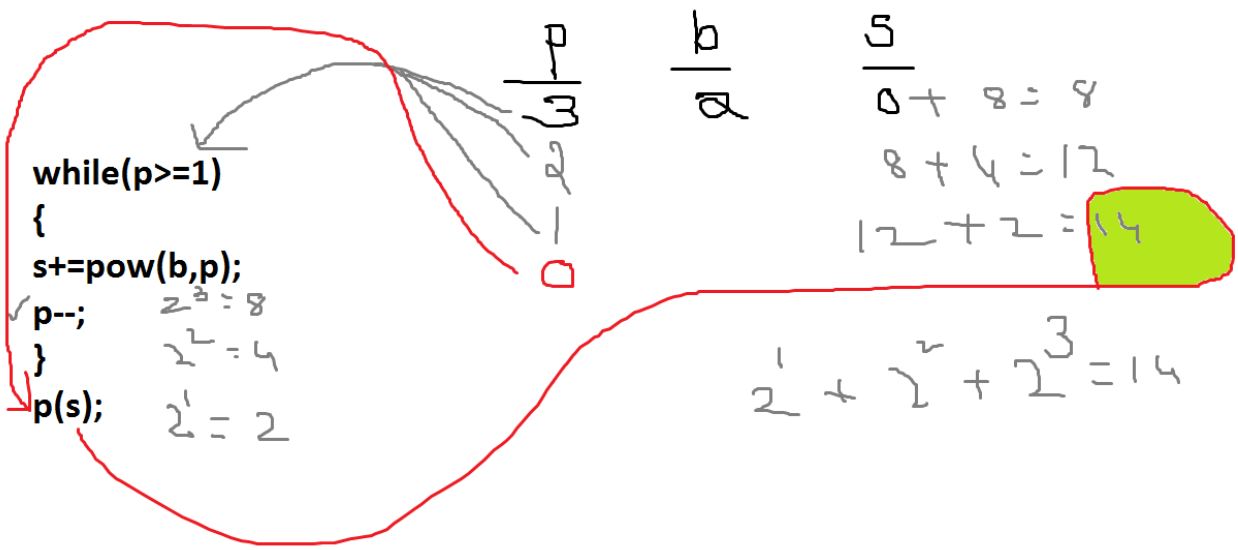
```
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int  b,p;
long s=0;
clrscr();
printf("Enter base, power values "); scanf("%d %d",&b, &p);
while(p>=1)
{
s+=pow(b,p);
p--;
}
printf("Powers sum=%ld", s);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

TC - exit

```
Enter base, power values 2 3
Powers sum=14
```

Activate Windows
Go to PC settings to activate Windows.



Home work:

Printing first and last digit of given no.

2048 ➔ 2 first digit, 8 last digit.

Finding no of digits in given no.

387 ➔ 3 digit no

Finding max and min digits of given no.

Eg: 29108 ➔ 9 max, 0 min